sps_sequence	[table]
visit_set_id	int4 not null
sequence_type	varchar
name	varchar
comments	varchar
cmd_str	varchar
status	varchar

guide_stars	[table]
guide_star_id	int8 not null
ra	float8(17, 17)
decl	float8(17, 17)
cat_id	int4
obj_type_id	int4
mag_agc	float4(8, 8)
flux_agc	float4(8, 8)
flags	int4
created_at	timestamp
updated_at	timestamp

input_catalog	[table]
cat_id	int4 not null
input_catalog_name	varchar
input_catalog_description	varchar
created_at	timestamp
updated_at	timestamp

fiducial_fiber	[table]
fiducial_fiber_id	int4 not null
field_on_pfi	int4
ff_in_field	int4
ff_type	varchar(5)
ff_id_in_type	int4
mask	int4
version	varchar

fiducial_fiber_geometry	[table]
fiducial_fiber_id	int4 not null
fiducial_fiber_calib_id	int4 not null
ff_center_on_pfi_x_mm	float4(8, 8)
ff_center_on_pfi_y_mm	float4(8, 8)
elevation	float4(8, 8)
ambient_temp	float4(8, 8)

fiducial_fiber_calib	[table]
fiducial_fiber_calib_id	serial not null
	auto-incremented
calibrated_at	timestamp
comments	varchar

cobra_motor_calib	[table]
cobra_motor_calib_id	serial not null
	auto-incremented
calibrated_at	timestamp
comments	varchar

cobra_motor_model	[table]
cobra_motor_model_id	serial not null
	auto-incremented
cobra_motor_calib_id	int4
cobra_id	int4
cobra_motor_axis_id	int4
cobra_motor_direction_id	int4
cobra_motor_on_time	float4(8, 8)
cobra_motor_step_size	float4(8, 8)
cobra_motor_frequency	float4(8, 8)

cobra_convergence_test	[table]
cobra_motor_model_id	int4 not null
iteration	int4 not null
cobra_motor_angle_target_id	int4 not null
cobra_motor_angle_target	float4(8, 8)
cobra_motor_angle_difference	float4(8, 8)
signal_to_noise_ratio	float4(8, 8)

cobra_motor_axis	[table]
cobra_motor_axis_id	int4 not null
cobra_motor_axis_name	varchar

cobra_motor_direction	[table]
cobra_motor_direction_id	int4 not null
cobra_motor_direction_name	varchar

cobra_motor_map	[table]
cobra_motor_model_id	int4 not null
cobra_motor_move_sequence	int4 not null
cobra_motor_angle	float4(8, 8)
cobra_motor_speed	float4(8, 8)

agc_data	[table]
agc_exposure_id	int4 not null
agc_camera_id	int4 not null
spot_id	int4 not null
image_moment_00_pix	float4(8, 8)
centroid_x_pix	float4(8, 8)
centroid_y_pix	float4(8, 8)
central_image_moment_11_pix	float4(8, 8)
central_image_moment_20_pix	float4(8, 8)
central_image_moment_02_pix	float4(8, 8)
peak_pixel_x_pix	int4
peak_pixel_y_pix	int4
peak_intensity	float4(8, 8)
background	float4(8, 8)
estimated_magnitude	float4(8, 8)
flags	int4

agc_match	[table]
agc_exposure_id	int4 not null
agc_camera_id	int4 not null
spot_id	int4 not null
pfs_design_id	int8
guide_star_id	int8
agc_nominal_x_mm	float4(8, 8)
agc_nominal_y_mm	float4(8, 8)
agc_center_x_mm	float4(8, 8)
agc_center_y_mm	float4(8, 8)
flags	int4
camera_enabled	bool

obs_condition	[table]
pfs_visit_id	int4 not null
status_sequence_id	int4 not null
airmass	float4(8, 8)
moon_phase	float4(8, 8)
moon_alt	float4(8, 8)
moon_sep	float4(8, 8)
seeing	float4(8, 8)
transparency	float4(8, 8)
cloud_condition_id	int4
created_at	timestamp

cloud_condition	[table]
cloud_condition_id	int4 not null
cloud_condition_name	varchar
cloud_condition_description	varchar
created_at	timestamp
updated_at	timestamp

agc_exposure	[table]
agc_exposure_id	int4 not null
pfs_visit_id	int4
agc_exptime	float4(8, 8)
altitude	float4(8, 8)
azimuth	float4(8, 8)
insrot	float4(8, 8)
adc_pa	float4(8, 8)
m2_pos3	float4(8, 8)
outside_temperature	float4(8, 8)
outside_pressure	float4(8, 8)
outside_humidity	float4(8, 8)
measurement_algorithm	varchar
version_actor	varchar
version_instdata	varchar
taken_at	timestamp

agc_guide_offset	[table]
agc_exposure_id	int4 not null
guide_ra	float8(17, 17)
guide_dec	float8(17, 17)
guide_pa	float4(8, 8)
guide_delta_ra	float4(8, 8)
guide_delta_dec	float4(8, 8)
guide_delta_insrot	float4(8, 8)
guide_delta_scale	float4(8, 8)
guide_delta_az	float4(8, 8)
guide_delta_el	float4(8, 8)
guide_delta_z	float4(8, 8)
guide_delta_z1	float4(8, 8)
guide_delta_z2	float4(8, 8)
guide_delta_z3	float4(8, 8)
guide_delta_z4	float4(8, 8)
guide_delta_z5	float4(8, 8)
guide_delta_z6	float4(8, 8)
mask	int4
taken_at	timestamp

beam_switch_mode	[table]
beam_switch_mode_id	int4 not null
beam_switch_mode_name	varchar
beam_switch_mode_description	varchar

sps_condition	[table]
pfs_visit_id	int4 not null
sps_camera_id	int4 not null
background	float4(8, 8)
throughput	float4(8, 8)

sps_annotation	[table]
annotation_id	serial not null
	auto-incremented
pfs_visit_id	int4
sps_camera_id	int4
data_flag	int4
notes	varchar
created_at	timestamp

sps_visit	[table]
pfs_visit_id	int4 not null
exp_type	varchar

sps_exposure	[table]
pfs_visit_id	int4 not null
sps_camera_id	int4 not null
exptime	float4(8, 8)
time_exp_start	timestamp
time_exp_end	timestamp
beam_config_date	float8(17, 17)

sps_camera	[table]
sps_camera_id	int4 not null
sps_camera_name	varchar(2)
sps_module_id	int4
sps_module_name	varchar(3)
arm	varchar(1)
arm_num	int4

sps_module	[table]
sps_module_id	int4 not null
description	varchar

cobra_target	[table]
pfs_visit_id	int4 not null
iteration	int4 not null
cobra_id	int4 not null
pfs_config_id	int8
pfi_nominal_x_mm	float4(8, 8)
pfi_nominal_y_mm	float4(8, 8)
pfi_target_x_mm	float4(8, 8)
pfi_target_y_mm	float4(8, 8)
flags	int4

cobra_match	[table]
pfs_visit_id	int4 not null
iteration	int4 not null
cobra_id	int4 not null
mcs_frame_id	int4
spot_id	int4
pfi_center_x_mm	float4(8, 8)
pfi_center_y_mm	float4(8, 8)
flags	int4

cobra_move	[table]
pfs_visit_id	int4 not null
iteration	int4 not null
cobra_id	int4 not null
cobra_motor_model_id_theta	int4
motor_target_theta	float4(8, 8)
motor_num_step_theta	int4
motor_on_time_theta	float4(8, 8)
cobra_motor_model_id_phi	int4
motor_target_phi	float4(8, 8)
motor_num_step_phi	int4
motor_on_time_phi	float4(8, 8)
flags	int4

obslog_visit_note	[table]
id	serial not null
	auto-incremented
user_id	int4
pfs_visit_id	int4
body	varchar not null

obslog_user	[table]
id	serial not null
	auto-incremented
account_name	varchar not null

obslog_mcs_exposure_note	[table]
id	serial not null
	auto-incremented
user_id	int4
mcs_exposure_frame_id	int4
body	varchar not null

fiducial_fiber_match	[table]
pfs_visit_id	int4 not null
iteration	int4 not null
fiducial_fiber_id	int4 not null
mcs_frame_id	int4
spot_id	int4
pfi_center_x_mm	float4(8, 8)
pfi_center_y_mm	float4(8, 8)
fiducial_tweaked_x_mm	float4(8, 8)
fiducial_tweaked_y_mm	float4(8, 8)
flags	int4
match_mask	int4

mcs_exposure	[table]
mcs_frame_id	int4 not null
pfs_visit_id	int4
mcs_exptime	float4(8, 8)
altitude	float4(8, 8)
azimuth	float4(8, 8)
insrot	float4(8, 8)
adc_pa	float4(8, 8)
dome_temperature	float4(8, 8)
dome_pressure	float4(8, 8)
dome_humidity	float4(8, 8)
outside_temperature	float4(8, 8)
outside_pressure	float4(8, 8)
outside_humidity	float4(8, 8)
mcs_cover_temperature	float4(8, 8)
mcs_m1_temperature	float4(8, 8)
mcs_camera_id	int4
measurement_algorithm	varchar
version_actor	varchar
version_instdata	varchar
version_pfs_utils	varchar
taken_at	timestamp
taken_in_hst_at	timestamp

mcs_data	[table]
mcs_frame_id	int4 not null
spot_id	int4 not null
mcs_center_x_pix	float4(8, 8)
mcs_center_y_pix	float4(8, 8)
mcs_second_moment_x_pix	float4(8, 8)
mcs_second_moment_y_pix	float4(8, 8)
mcs_second_moment_xy_pix	float4(8, 8)
bgvalue	float4(8, 8)
peakvalue	float4(8, 8)
flags	int4
flux	float4(8, 8)
fluxerr	float4(8, 8)

mcs_camera	[table]
mcs_camera_id	int4 not null
mcs_camera_name	varchar
comments	varchar

mcs_pfi_transformation	[table]
mcs_frame_id	int4 not null
x0	float4(8, 8)
y0	float4(8, 8)
dscale	float4(8, 8)
scale2	float4(8, 8)
theta	float4(8, 8)
alpha_rot	float4(8, 8)
camera_name	varchar
mcs_boresight_x_pix	float4(8, 8)
mcs_boresight_y_pix	float4(8, 8)

tel_status	[table]
pfs_visit_id	int4 not null
status_sequence_id	int4 not null
altitude	float4(8, 8)
azimuth	float4(8, 8)
insrot	float4(8, 8)
inst_pa	float4(8, 8)
adc_pa	float4(8, 8)
m2_pos3	float4(8, 8)
m2_off3	float4(8, 8)
tel_ra	float4(8, 8)
tel_dec	float4(8, 8)
dither_ra	float4(8, 8)
dither_dec	float4(8, 8)
dither_pa	float4(8, 8)
dome_shutter_status	int4
dome_light_status	int4
caller	varchar
created_at	timestamp

camera_model_f3c_mcs	[table]
mcs_frame_id	int4 not null
matrix	_float4
distor	_float4
rot_vector	_float4
tran_vector	_float4

obslog_visit_set_note	[table]
id	serial not null
	auto-incremented
user_id	int4
iic_sequence_id	int4
body	varchar not null

mcs_boresight_frame_set	[table]
mcs_boresight_frame_set_id	int4 not null
mcs_frame_id	int4 not null

env_condition	[table]
pfs_visit_id	int4 not null
status_sequence_id	int4 not null
dome_temperature	float4(8, 8)
dome_pressure	float4(8, 8)
dome_humidity	float4(8, 8)
outside_temperature	float4(8, 8)
outside_pressure	float4(8, 8)
outside_humidity	float4(8, 8)
created_at	timestamp

pfs_visit	[table]
pfs_visit_id	int4 not null
pfs_visit_description	varchar
pfs_design_id	int8
issued_at	timestamp

mcs_boresight	[table]
pfs_visit_id	int4 not null
mcs_boresight_x_pix	float4(8, 8)
mcs_boresight_y_pix	float4(8, 8)
calculated_at	timestamp

mcs_boresight_meas	[table]
pfs_visit_id	int4 not null
mcs_boresight_x_pix	float4(8, 8)
mcs_boresight_y_pix	float4(8, 8)
calculated_at	timestamptz not null
mcs_boresight_frame_set_id	int4
mcs_boresight_x_err_pix	float4(8, 8)
mcs_boresight_y_err_pix	float4(8, 8)
altitude	float4(8, 8)

obslog_fits_header	[table]
pfs_visit_id	int4
filestem	varchar not null
hdu_index	int4 not null
cards_dict	jsonb not null
cards_list	jsonb not null

pfs_config_sps_observed	[table]
pfs_visit_id	int4 not null
observed_at	timestamptz

visit_set	[table]
pfs_visit_id	int4 not null
iic_sequence_id	int4

iic_sequence	[table]
iic_sequence_id	int4 not null
sequence_type	varchar
name	varchar
comments	varchar
cmd_str	varchar
group_id	int4
created_at	timestamp

iic_sequence_status	[table]
iic_sequence_id	int4 not null
status_flag	int4
cmd_output	varchar
finished_at	timestamp

sequence_group	[table]
group_id	int4 not null
group_name	varchar
created_at	timestamp

dot_roach_flux	[table]
pfs_visit_id	int4 not null
cobra_id	int4 not null
flux_abs	float4(8, 8)
flux_ratio	float4(8, 8)
flux_ratio_norm	float4(8, 8)

cobra_geometry_calib	[table]
cobra_geometry_calib_id	serial not null
	auto-incremented
calibrated_at	timestamp
comments	varchar

cobra_geometry	[table]
cobra_geometry_calib_id	int4 not null
cobra_id	int4 not null
center_x_mm	float4(8, 8)
center_y_mm	float4(8, 8)
motor_theta_limit0	float4(8, 8)
motor_theta_limit1	float4(8, 8)
motor_theta_length_mm	float4(8, 8)
motor_phi_limit_in	float4(8, 8)
motor_phi_limit_out	float4(8, 8)
motor_phi_length_mm	float4(8, 8)
status	int4

field_set	[table]
iic_sequence_id	int4 not null
visit0	int4

cobra	[table]
cobra_id	int4 not null
field_on_pfi	int4
cobra_in_field	int4
cobra_module_id	int4
module_in_field	int4
cobra_in_module	int4
cobra_in_board	int4
cobra_board_id	int4
mtp	varchar(3)
sps_module_id	int4
sps_slit_hole	int4
science_fiber_id	int4
fiber_id	int4
sunss_id	varchar(4)
mtp_a_id	varchar
mtp_c_id	varchar
mtp_ba_id	varchar
mtp_bc_id	varchar
version	varchar
created_at	timestamp
updated_at	timestamp

cobra_blind_move	[table]
pfs_visit_id	int4 not null
cobra_id	int4 not null
dot_landing	float4(8, 8)
dot_target	float4(8, 8)
dot_command	float4(8, 8)
dot_approach	float4(8, 8)
phi_landing_target	float4(8, 8)
phi_push_target	float4(8, 8)
phi_start	float4(8, 8)
phi_landed	float4(8, 8)
phi_pushed	float4(8, 8)
ramp_residual	float4(8, 8)
phi_landing_command	float4(8, 8)
phi_push_start	float4(8, 8)
phi_push_command	float4(8, 8)
blind_move_calib_id	int4

pfs_config_sps	[table]
pfs_visit_id	int4 not null
visit0	int4 not null
cam_mask	int4
inst_status_flag	int4
requested_at	timestamptz

cobra_blind_move_gain	[table]
blind_move_calib_id	int4 not null
cobra_id	int4 not null
landing_gain	float4(8, 8)
target_gain	float4(8, 8)
n_runs	int4
landing_scatter	float4(8, 8)
target_scatter	float4(8, 8)
calibrated	bool not null
reason	varchar

blind_move_calib	[table]
blind_move_calib_id	serial not null
	auto-incremented
created_at	timestamptz not null
visits	_int4 not null
min_runs	int4 not null
model	varchar not null
model_version	int4 not null
n_calibrated	int4 not null
loo_bias	float4(8, 8)
loo_spread	float4(8, 8)
loo_within_05	float4(8, 8)
prior_calib_id	int4
dot_landing	float4(8, 8)
dot_target	float4(8, 8)
parameters	varchar

blind_move_calib_validation	[table]
blind_move_calib_validation_id	serial not null
	auto-incremented
blind_move_calib_id	int4 not null
validated_at	timestamptz not null
visits	_int4 not null
n_cobras	int4
bias	float4(8, 8)
spread	float4(8, 8)
within_05	float4(8, 8)
within_10	float4(8, 8)
coverage	float4(8, 8)
reason	varchar not null

pfs_config_agc	[table]
pfs_design_id	int8 not null
visit0	int4 not null
guide_star_id	int8 not null
agc_camera_id	int4
agc_final_x_pix	float4(8, 8)
agc_final_y_pix	float4(8, 8)
comments	varchar
guide_star_ra	float8(17, 17)
guide_star_dec	float8(17, 17)

pfs_config	[table]
pfs_design_id	int8 not null
visit0	int4 not null
ra_center_config	float8(17, 17)
dec_center_config	float8(17, 17)
pa_config	float4(8, 8)
converg_num_iter	int4
converg_elapsed_time	float4(8, 8)
converg_tolerance	float4(8, 8)
alloc_rms_scatter	float4(8, 8)
allocated_at	timestamp
to_be_observed_at	timestamptz
pfs_utils_version	varchar
to_be_observed_at_design	timestamptz
pfs_utils_version_design	varchar
was_observed	bool
converg_distance_threshold	float4(8, 8)
target_fallback_invalid	varchar
target_fallback_unassigned	varchar
fiducial_check_skipped	bool
inst_status_flag	int4
blind_move_calib_id	int4
dot_landing	float4(8, 8)
dot_target	float4(8, 8)

pfs_config_fiber	[table]
pfs_design_id	int8 not null
visit0	int4 not null
fiber_id	int4 not null
target_ra	float8(17, 17)
target_dec	float8(17, 17)
fiber_status	int4
target_validation_mask	int4
cobra_command	int4
pfi_nominal_x_mm	float4(8, 8)
pfi_nominal_y_mm	float4(8, 8)
pfi_center_final_x_mm	float4(8, 8)
pfi_center_final_y_mm	float4(8, 8)
motor_map_summary	varchar
config_elapsed_time	float4(8, 8)
is_on_source	bool
comments	varchar

pfs_design_fiber	[table]
pfs_design_id	int8 not null
fiber_id	int4 not null
target_cat_id	int4
target_tract	int4
target_patch	varchar
target_obj_id	int8
target_ra	float8(17, 17)
target_dec	float8(17, 17)
target_pm_ra	float4(8, 8)
target_pm_dec	float4(8, 8)
target_parallax	float4(8, 8)
epoch	varchar
proposal_id	varchar
ob_code	varchar
target_type	int4
fiber_status	int4
pfi_nominal_x_mm	float4(8, 8)
pfi_nominal_y_mm	float4(8, 8)
ets_priority	int4
ets_cost_function	float8(17, 17)
ets_cobra_motor_movement	varchar
is_on_source	bool
comments	varchar

pfs_design	[table]
pfs_design_id	int8 not null
design_name	varchar
variant	int4 not null
design_id0	int8 not null
tile_id	int4
ra_center_designed	float8(17, 17)
dec_center_designed	float8(17, 17)
pa_designed	float4(8, 8)
num_sci_designed	int4
num_cal_designed	int4
num_sky_designed	int4
num_guide_stars	int4
exptime_tot	float4(8, 8)
exptime_min	float4(8, 8)
ets_version	varchar
ets_assigner	varchar
designed_at	timestamp
to_be_observed_at	timestamptz
pfs_utils_version	varchar
is_obsolete	bool

calib_set	[table]
calib_set_id	serial not null
	auto-incremented
calib_flat_id	int4
calib_bias_id	int4
calib_dark_id	int4
calib_arcs_id	int4

calib	[table]
calib_id	bigserial not null
	auto-incremented
calib_type	varchar
sps_frames_to_use	varchar
pfs_design_id	int8
calib_date	timestamp

pfs_design_agc	[table]
pfs_design_id	int8 not null
guide_star_id	int8 not null
epoch	varchar
guide_star_ra	float8(17, 17)
guide_star_dec	float8(17, 17)
guide_star_pm_ra	float4(8, 8)
guide_star_pm_dec	float4(8, 8)
guide_star_parallax	float4(8, 8)
guide_star_magnitude	float4(8, 8)
passband	varchar
guide_star_color	float4(8, 8)
guide_star_flag	int4
agc_camera_id	int4
agc_target_x_pix	float4(8, 8)
agc_target_y_pix	float4(8, 8)
comments	varchar

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